

IO SKY

AI SALES OUTBOUND

Functional & Technical Implementation Specification

Version 1.0 - Final Developer Handoff

Document owner	IO SKY
System	IO SKY Operational Intelligence Platform
Capability	AI Sales Outbound
Audience	Development, Architecture, QA, Security, Operations
Status	Final Product Owner Functional Direction
Implementation rule	Requirements may not be silently omitted, weakened or reinterpreted.

Research before outreach. Evidence before assumption. Qualification before automation.
Autonomy exists within authority.

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1. Product Definition & Objective

AI Sales Outbound shall be implemented as an intelligent, autonomous and human-controllable commercial prospecting and outbound capability within the IO SKY platform. It is not merely an email generator, lead scraper, vacancy bot, bulk-mail tool, CRM add-on or chatbot.

The canonical commercial lifecycle is: Discovery -> Signal Detection -> Organisation Resolution -> Relationship Protection -> Research -> Evidence -> Qualification -> Contact Discovery -> Opportunity -> Outreach Strategy -> Personalisation -> Approval / Authorised Autonomy -> Execution -> Follow-up -> Reply Handling -> Scheduling -> Commercial Administration -> Controlled Learning.

ASO-CORE-001 AI Sales Outbound shall automate commercial execution without automating away IO SKY's control.

ASO-CORE-002 Every automated workflow must leave IO SKY's commercial administration more complete and organised, never more fragmented.

ASO-CORE-003 The objective is higher-quality commercial opportunities with substantially less manual work, not maximum outreach volume.

2. System Architecture & Authority

AI Sales Outbound shall reuse authoritative platform services rather than create parallel systems. Shared services include Identity and authentication, RBAC/permissions, Organisation and Person lifecycle, global search, Notification Center, Scheduling Engine, telephony infrastructure, audit infrastructure, secrets/configuration, analytics/events and approved communication infrastructure.

ASO-ARC-001 Critical business state shall be stored outside the LLM in authoritative platform/database services.

ASO-ARC-002 Where AI reasoning conflicts with authoritative platform state, authoritative platform state wins.

3. Authority Chain

Every consequential action shall require a valid authority chain: Actor/Agent -> Permission -> Scope -> Campaign -> Campaign Mandate -> Action Type -> Policy -> Resource -> Current State.

Capability means technically possible; Permission means the actor is authorised; Mandate means the campaign permits it; Policy means applicable rules permit it; State means the current workflow/relationship still permits it.

ASO-AUT-001 Failure of any required authority condition shall block execution.

ASO-AUT-002 The AI may never grant itself additional authority.

ASO-AUT-003 Model intent is not permission; backend validation remains authoritative.

4. Operating Modes

- Research Only - discovery, research, evidence, qualification, recommendations and drafting; no external communication.
- Approval Mode - consequential external actions require authorised human approval before execution.
- Autonomous Mode - the system may independently perform specifically authorised actions inside an explicit Campaign Mandate.

ASO-AUT-004 Autonomous Mode means independent execution inside an explicitly controlled authority envelope; it never means unrestricted AI authority.

5. Campaign Mandate

Every autonomous campaign shall define its ICP/target, geography, offer/service boundaries, permitted sources and channels, sending identities, qualification requirements, exclusions, communication policy, approved claim boundaries, contact-frequency and contact-time rules where applicable, volume/rate limits, follow-up policy, escalation requirements, commercial authority and applicable model/cost limits.

Every consequential action shall resolve to Automatic, Approval Required or Prohibited. Effective authority may vary by campaign, channel, geography, relationship, signal, action or risk category.

6. Campaign Lifecycle

Campaigns shall support Draft -> Ready -> Active -> Paused -> Completed and, where applicable, Stopped / Cancelled / Failed / Archived.

Pause stops new autonomous external actions while retaining state. Stop terminates future campaign execution. Kill Switch immediately prevents new consequential Sales Outbound actions within its configured scope.

ASO-AUT-005 Resume shall never blindly execute stale queued actions; current-state validation must occur first.

7. Pre-Action Revalidation

Immediately before every consequential action, the backend shall revalidate relevant authority and current state, including permission, campaign state and mandate, relationship, suppression/DNC, collision, contact frequency, contact validity, sending identity, channel availability, rate/volume policy, connector status and approval validity.

ASO-AUT-006 A previously valid decision does not override newer authoritative state. Example: approved draft -> unsubscribe -> queued send must be blocked.

8. Core Data Model

Organisation, Person, Signal, Evidence, Opportunity, Campaign, Campaign Membership, Interaction, Approval and Human Attention Item shall be separate first-class concepts. They shall not be collapsed into one generic Lead object.

9. Organisation

One Organisation may contain multiple People, Signals, Opportunities, Campaign Memberships, Interactions, meetings, projects, AI Scans and customer history.

One organisation, one coherent commercial history, multiple evidence-backed signals and opportunities.

10. Organisation Identity Resolution

Matching shall consider reliable identifiers such as canonical domain, known email domains, company/CRM/source identifiers, aliases/trading names and canonical URLs where available.

ASO-DATA-001 Materially ambiguous identity shall not be blindly merged; route to review/Human Attention where necessary.

11. Person

A Person is independent of an Opportunity, may belong to an Organisation, participate in several opportunities, appear in historical interactions, change roles and have multiple contact methods. The same individual shall not be duplicated merely because another source discovers them.

12. Signal

A Signal is an observed circumstance potentially relevant to a commercial opportunity, such as a vacancy, contractor requirement, technology initiative, expansion or other authorised observable business event.

ASO-DATA-002 A Signal is not proof that IO SKY is needed; it triggers investigation.

13. Evidence

Evidence shall retain source, URL/source identifier, observed/retrieved date, summary, freshness, associated Signal/Opportunity and provenance where available. The platform shall distinguish Verified Fact, Evidence, Inference/Hypothesis and Unknown.

ASO-RES-001 Verified Fact, Evidence, Inference/Hypothesis and Unknown may never be silently collapsed.

14. Opportunity

Opportunity is a first-class commercial object linking Organisation, relevant Person(s), Signal(s), Evidence, potential IO SKY relevance, qualification, source attribution, Campaign, outreach history, replies, Scheduling activity, Discovery Call, commercial progression, status, next action and owner/assignment where applicable. A new Signal may create or enrich an Opportunity; it does not automatically create a new Organisation or Person.

15. Opportunity Lifecycle

The implementation shall distinguish concepts equivalent to Detected -> Researching -> Qualified / Not Qualified -> Outreach Ready -> Contacted -> Engaged -> Meeting Intent -> Meeting Scheduled -> Commercial Follow-up -> Converted / Closed / On Hold, with additional states for Insufficient Evidence, Human Review Required, Closed Lost and No Current Opportunity. Technical enum names may differ; materially different business states may not be collapsed.

16. Relationship State

Organisation relationship state is separate from Opportunity state. Relevant concepts include New/Unknown Prospect, Active Prospect, Existing Client, Former Client and Partner/Other Relationship. Do Not Contact is not a relationship state; it belongs to the suppression/compliance layer.

17. Existing Client Protection

Before cold qualification/execution: Signal -> Organisation Resolution -> Relationship Check -> Suppression Check -> Opportunity Routing -> Research/Qualification. Existing clients shall never enter ordinary cold-prospecting workflows. A relevant new Signal for an Existing Client shall create an Existing Client Opportunity Alert containing client, Signal, Evidence, relevance, relationship context, potential opportunity and recommended next action. Default handling is relationship-managed/personal follow-up.

18. Existing Prospect & Former Client Protection

Active Prospects retain prior history. Former Clients remain separately identifiable. A materially new Signal may justify re-engagement only after prior history, the new Signal, suppression, frequency rules and justification are evaluated.

19. Campaign Collision Protection

Immediately before outreach, check Person, Organisation, active Campaign Memberships, queued/recent outreach, scheduled follow-up, open Opportunity, existing-client relationship and suppression. Collision shall cause Block, Defer, Merge, Reroute or Human Review according to policy.

20. Contact Frequency Guard

Frequency controls shall operate independently of model judgment at Person and Organisation level. AI may recommend renewed contact but cannot bypass the configured guard. Authorised overrides, where allowed, must be explicit and audited.

21. Discovery & Source Architecture

The architecture shall support at least 10 configurable source connectors without architectural redesign. Intended launch mix is approximately five useful sources/categories: Indeed; Werken voor Nederland/relevant Dutch governmental vacancy environments; one appropriate Dutch commercial vacancy source; company career pages; and one appropriate freelance/contract opportunity source. Exact providers remain subject to real access, API/integration availability, applicable terms and production feasibility.

22. Source Orchestrator

The Source Orchestrator shall intelligently select sources using factors such as ICP, geography, Campaign, Signal type, historical yield, freshness, coverage, duplication, cost, provider limits, previous search coverage and availability. It shall not mechanically query every source for every task. Controlled Learning may improve source selection.

23. Connector Capability Matrix

Every production connector shall document whether it supports Discover, Search, Read, Research, Retrieve Detail, Qualify, Draft, Direct Submit, Direct Send, Read Reply, Reply and Retrieve Status, with states such as Supported, Unsupported, Approval Required, Autonomous Allowed or Human Action Required. Each launch connector shall document integration method, authentication, capabilities, limitations, rate limits, data retrieved/stored, approval requirements, error behaviour and production prerequisites.

ASO-CON-001 A visible source logo/card is not a completed integration.

24. Integration Priority

Preferred integration order: Official API/authorised integration -> permitted structured/public access -> permitted browser/computer interaction -> Human Action Required. Browser/computer interaction is an adapter/fallback, not the core architecture. The implementation shall not circumvent access controls or anti-automation safeguards, and IO SKY shall not be falsely represented as a job applicant.

25. Vacancy Intelligence

A vacancy is a commercial Signal, not proof of need. Verified hiring activity may justify investigation; it does not justify invented claims about operational failure or pain.

26. Employment-Type Intelligence

Where evidence permits, distinguish permanent employment, freelance, contractor, temporary, project engagement and other relevant structures. Permanent-role outreach may position IO SKY as an alternative/complement to hiring where credible; freelance/contract outreach may more directly address specialist/project capacity. No absolute guarantees may be invented.

ASO-RES-002 Permanent-employment and freelance/contract situations shall not simply receive the same template with different job titles.

27. Discovery Source vs Outreach Channel

Discovery source and communication channel are independent. A Signal discovered on a vacancy source may lead to authorised business email or another permitted route. Direct source-platform execution occurs only if the connector supports and permits it.

28. Manual Research

An authorised user shall be able to provide a company, domain, URL, person, vacancy, opportunity or other supported reference and instruct the agent to research it. Natural-language instructions shall be supported where practical.

29. Bulk Import

Launch shall support CSV and XLSX with no artificial 50-record ceiling. Hundreds of records shall be processable through durable background workflows. Canonical pipeline: Upload -> Validate -> Normalise ->

Deduplicate -> Organisation Resolution -> CRM Match -> Relationship Check -> Research -> Evidence -> Qualification -> Recommended Action. Import itself does not authorise sending.

30. Bulk Import States

Relevant results include Qualified, Potential Fit, Needs Review, Disqualified, Insufficient Evidence, Existing Client, Existing Active Prospect, Duplicate and Do Not Contact/Suppressed where applicable. Large imports require progress, retry, pause/resume and recoverability.

31. Research Brief

Before material personalised outreach, build a Research Brief containing where available: organisation summary, observed Signal, published requirement, relevant developments, hiring/technology/operational signals, relevant Person/function, potential IO SKY relevance, Evidence, Unknowns, Inferences, recommended approach, evidence dates/sources and Why this message?. Research depth shall scale with opportunity value, uncertainty and policy.

32. Research Integrity

ASO-RES-003 No invented problems: hypotheses may not become facts.

ASO-RES-004 Time-sensitive evidence shall be freshness-aware.

ASO-RES-005 Conflicting evidence shall trigger further research, stronger-model escalation, unresolved state or Human Attention - not fabricated certainty.

ASO-RES-006 Research failure shall produce Research Incomplete / Insufficient Evidence, not Disqualified unless independent evidence justifies disqualification.

33. Qualification Engine

Qualification shall combine ICP, verified Signals, Evidence, Organisation context, geography, service relevance, campaign requirements, exclusions, relationship state and evidence quality as appropriate. It shall not be simple keyword matching.

34. ICP Configuration

Multiple ICPs shall be configurable without code changes. Criteria may include organisation type, size, sector, geography, maturity, Signals, technology context, hiring activity, operational context, commercial characteristics, service relevance and exclusions. Architecture shall support Dutch and international ICPs.

35. Explainable Qualification

Any score must be decomposable into criteria, Evidence, positive/negative factors, uncertainty and missing information. A black-box 87/100 is insufficient. Authorised Super Admin override shall preserve the original AI assessment and Evidence and create an audit event.

36. Service Relevance

Where evidence supports it, relevance may include Engineering / Custom Software, AI & Automation, Data Infrastructure, Cloud / Infrastructure, Security, recurring IO SKY systems and other approved capabilities.

ASO-QUAL-001 The existence of an IO SKY service is not evidence that a prospect needs it.

37. Contact Discovery

After sufficient qualification, the system may identify relevant business contacts using authorised methods. It shall explain why a selected contact/function is appropriate. Contact data should retain source, verification/confidence and retrieval date where practical. Guessed data shall not be represented as verified.

38. Suppression / Do Not Contact

Suppression is a central control layer across campaigns/channels and includes Do Not Contact, unsubscribe, hard suppression and other configured prohibited-contact conditions. Suppression shall be rechecked immediately before external execution. Learning, Campaign configuration and model reasoning may never bypass it.

39. IO SKY Sales Communication Profile

The centrally governed communication profile shall produce communication that is short, powerful, specific, professional, human, confident and evidence-led. Avoid generic AI prose, unnecessary length, fake compliments, invented pain, exaggerated familiarity, unsupported claims, desperation, generic vacancy templates and premature solution prescription. Preferred logic: Relevant observation -> credible reason for contact -> relevant IO SKY capability -> simple next step.

40. Personalisation Standard

Personalisation is not first name + company + vacancy title + template. Before material outreach, the agent should understand the Organisation, actual Signal, Evidence, likely relevance, appropriate Person, relevant IO SKY capability and remaining Unknowns. Every proposed personalised communication shall internally include Why this message? explaining Evidence, reasoning, angle, contact/channel rationale and material uncertainty.

41. Subject-Line Intelligence

The agent may generate multiple subject candidates and recommend one. Subjects shall be concise, specific, relevant and professional. Fake Re:, fake Fwd:, fabricated urgency or wording falsely implying an existing relationship are prohibited.

42. Approval Object

An Approval item shall show Organisation, source, Signal, Evidence, qualification, Person/contact, recommended channel, sending identity, subject, complete proposed communication, Why This Message, material uncertainty and intended action. Lifecycle: Pending -> Approved / Edited & Approved / Rejected / Expired / Invalidated.

43. Approval Invalidation

Approval shall be invalidated when a material underlying condition changes before execution, including material message regeneration, contact change, suppression, campaign pause/stop, material relationship change, sending identity change or applicable policy change.

44. Approval vs Execution

Approval and execution are distinct. Canonical flow: Draft -> Approved -> Queued -> Executing -> Provider/Connector Accepted -> Sent/Submitted; failure may instead become Execution Failed. Approve & Send shall actually execute through the authorised connector where supported after validation.

45. Autonomous Decision Trace

For material autonomous actions, retain enough structured information to determine what happened, which Campaign authorised it, which policy/rule allowed it, relevant Evidence, applicable model/task, backend validation result, connector and outcome. Private chain-of-thought is not required; operational reasoning summaries and decision metadata are.

46. Email Outbound Infrastructure

The intended dedicated outbound domain is iosky.co, becoming production-authoritative only after ownership, DNS and production configuration are verified. It shall remain configurable, not hardcoded. iosky.nl remains the core corporate/operational domain; iosky.io is reserved for future international brand positioning and shall not be the cold-outbound domain. Architecture: Outbound Domain -> Sending Identities/Mailboxes -> Sending Provider -> Deliverability Layer -> Sales Outbound Engine. Controls include SPF, DKIM, DMARC, bounce processing, suppression, unsubscribe handling where applicable, delivery status, sending limits and health/reputation visibility. No domain rotation to evade controls.

47. Sending Identity

The AI shall never invent a sender identity. Every external communication shall use an authorised configured identity. Approval Mode shall show the intended sender before approval; Autonomous Campaigns shall explicitly define permitted identities.

48. Channel Execution

Channels may include business email, authorised vacancy/source-platform communication, authorised voice and future approved channels. Capabilities are channel-specific. If direct execution is unavailable, create Human Action Required with destination, context, proposed communication, Evidence and reason manual execution is required.

49. Follow-up Engine

Follow-up is launch functionality. Cadence is configurable by Campaign/channel. Follow-ups use prior context and shall not merely paraphrase the initial message. A genuine reply interrupts the no-reply sequence. Stop/transition conditions include reply, unsubscribe/DNC, hard bounce, permanent failure, definitive rejection, meeting booked, relationship change and Campaign stop as applicable.

50. Reply Intelligence

At minimum distinguish Interested, More Information, Referral, Objection, Not Now, Not Relevant, Already Solved, Wrong Contact, Unsubscribe/DNC, Out of Office, Meeting Intent and Human Attention Required. Classification alone is insufficient; determine the next authorised action. Reliable OOO return dates may schedule an appropriate future action. Not Now may create a bounded future follow-up where policy permits, never endless nurture.

51. Reply Authority

Routine questions within approved IO SKY sales knowledge may be answered autonomously where the Campaign Mandate permits. Material bespoke pricing, contractual commitments, legal issues, security representations, architecture, guarantees and unusual commercial commitments shall escalate rather than be improvised.

52. Scheduling Integration

Meeting intent shall hand off to the authoritative Scheduling Agent; Sales Outbound shall not create a second availability/booking engine. Flow: Meeting Intent -> Scheduling Agent -> Authoritative Availability -> Booking -> Confirmation -> Lifecycle Update. A prospect request such as Call me Friday may create an authorised Scheduling Task containing Sales Outbound context.

53. AI Receptionist & Telephony Integration

AI Sales Outbound is proactive commercial outreach; AI Receptionist is inbound first point of contact; Scheduling Agent is authoritative scheduling capability. They may share authorised lifecycle context, but shared infrastructure does not mean shared authority. Context is intelligence, not narration. Shared telephony may be reused where authorised; Sales Outbound shall not duplicate the IVR/telephony foundation.

54. Commercial Lifecycle Administration

Successful workflows shall automatically maintain structured Organisation, Person, Signal, Evidence, Opportunity, Interaction, Campaign, response, Scheduling handoff, next action, source attribution, commercial outcome and audit records as applicable. Humans should not need to recreate basic AI-generated commercial administration manually.

55. Global Search & Filtering

Sales Outbound shall participate in permission-aware global search across relevant company, Person, domain, email, phone, Signal, vacancy, source, Campaign, Opportunity, service relevance, booking and status data. Filters shall work individually and in useful combinations. Searchability never expands access rights.

56. International & Language Architecture

Campaigns shall support Netherlands, selected countries/regions and international targeting. Launch communication languages are Dutch and English; additional languages shall be possible without redesign.

Professional localisation is required and facts, uncertainty, business meaning and IO SKY positioning must remain consistent. Raw literal machine translation is not an acceptable publication-quality standard.

57. Model Provider Architecture

OpenAI is the intended primary launch AI provider, but the architecture shall provide provider/model abstraction. Business Task -> Logical Capability Requirement -> Model Router -> Configured Provider/Model. Business logic shall not be distributed across hardcoded model identifiers.

58. Initial Model Routing Baseline

These are initial mappings, not permanent hardcoded rules.

- Luna - high-volume/lower-complexity extraction, normalisation, straightforward classification, tagging, preprocessing and low-complexity structured transformations.
- Terra - standard research synthesis, qualification, prospect analysis, standard personalised drafts, subject lines and routine reply reasoning.
- Sol - difficult research, conflicting Evidence, high-value accounts, difficult qualification, sophisticated personalisation and complex commercial replies.
- Astra - exceptional multi-step complexity, unusually difficult/high-value analysis and cases unresolved by permitted lower tiers.

59. Model Routing & Escalation

Conceptual escalation is Luna -> Terra -> Sol -> exceptional Astra, but not every task traverses every tier. The router shall seek the lowest-cost model reasonably capable of satisfying required quality. Escalation criteria may include insufficient confidence, evidence conflict, unresolved ambiguity, complexity, commercial importance, elevated risk, schema/quality failure or previous unsuccessful reasoning. If the highest permitted tier remains unresolved, use Human Attention / Insufficient Evidence / Research Incomplete. No model may invent certainty to finish a workflow.

60. Structured Relay / Handoff Contract

Escalation shall use structured state with fields equivalent to task_id, workflow_id, organisation_id, person_id where applicable, opportunity_id, campaign_id, task_type, objective, completed_steps, verified_facts, evidence_references, inferences, unknowns, conflicts, previous_output where appropriate, unresolved_questions, escalation_reason, previous_model and requested_next_capability. The stronger model continues from useful completed work. Sending an uncontrolled full conversation transcript is not a substitute.

61. Intelligence Escalation vs Technical Fallback

Intelligence Escalation means the model works technically but cannot reach required reliability, so route to a stronger permitted tier or Human Attention. Technical Fallback covers timeout, outage, rate limit, malformed response or infrastructure failure, using controlled retry/provider fallback. Technical failure is not evidence that stronger reasoning is required.

62. Model Loop Protection

Escalation/retry shall be bounded. Infinite loops such as Terra -> Sol -> Terra -> Sol are prohibited. Maximum retry/escalation policy shall be configurable; unresolved work terminates safely into a human/incomplete state.

63. Model Observability & Cost Governance

Track task type, initial model, escalation and reason, final model, technical fallback, latency, token/usage consumption, cost, status and outcome/quality signals as appropriate. Support Campaign and model/task cost visibility, daily/monthly budget controls or alerts and abnormal-usage visibility. Expensive escalation shall be justified by task need.

64. Model Version Resilience

Model/task mapping shall be centrally configurable. Models may be replaced when better capability becomes available, a model is deprecated, pricing/reliability changes or another authorised provider becomes preferable. No business workflow shall require reconstruction merely because the underlying model changes.

65. Controlled Learning Engine

Learning signals may include Approve, Edit, Reject, Regenerate, structured edit/rejection reasons, replies, positive replies, objections, meetings, opportunities, customers/commercial outcomes, source performance and Campaign performance. The system should improve prospect relevance, research depth, source selection, qualification, communication, model routing and efficiency.

66. Learning Hierarchy

Learning scope: Global IO SKY Rules -> Approved Sales Policy -> ICP/Campaign Learning -> Channel Learning -> Signal-Type Learning -> Organisation/Prospect History. A pattern learned in one context shall not blindly become global policy. Explicit authorised instruction outranks inferred learning.

67. Learning Boundaries

Learning shall never autonomously relax or rewrite suppression, permissions, Campaign Mandates, contact-frequency controls, compliance rules, security requirements, pricing authority, contractual authority, approved claims or fundamental IO SKY communication policy.

68. Learning Explainability

Authorised users shall be able to understand material learned preferences/patterns and, where appropriate, correct, disable, reset, version or roll back relevant behaviour. Structured feedback reasons should be captured where practical, such as too generic, unsupported assumption, too long, wrong contact, wrong angle, weak evidence, wrong service relevance or tone mismatch.

69. Learning Objective

Primary optimisation should favour relevant qualified replies, meetings, opportunities, commercial progression, customers/revenue where attributable, qualification quality, source/research efficiency and appropriate model cost. Opens/clicks may be measured but are not the primary objective.

70. Human Attention Queue

Human Attention is a first-class operational queue for Existing Client Opportunity, ambiguous Organisation, high-value review, complex reply, commercial exception, manual connector action, unresolved evidence, unusual request, policy uncertainty and repeated failure. Each item shall support reason, priority, originating workflow, Organisation, Person where applicable, Opportunity, context/Evidence, recommended action, assignment, creation time, due/attention time where relevant, status, resolution and audit history.

71. Permissions

Platform rule: Role + Permission + Scope + Assignment/Ownership + Environment + Resource. Frontend visibility never grants backend access. Super Admin has full authorised Sales Outbound control; Admin only granular granted permissions; Technical Operator only explicit technical scopes/resources; Developer no production prospect/client data by default.

72. Canonical Permission Coverage

Naming may follow canonical platform conventions; required granularity may not be removed.

- outbound.view
- outbound.research
- outbound.import
- outbound.qualification.view
- outbound.qualification.override
- outbound.campaign.view
- outbound.campaign.manage
- outbound.draft.view
- outbound.draft.edit
- outbound.draft.approve
- outbound.send
- outbound.autonomy.manage
- outbound.reply.view
- outbound.reply.manage
- outbound.suppression.manage
- outbound.connector.manage
- outbound.source.manage
- outbound.sending.manage
- outbound.model_routing.manage
- outbound.learning.view
- outbound.learning.manage
- outbound.analytics.view
- outbound.audit.view
- outbound.configuration.manage

73. Auditability

Material Campaign configuration, approvals, edits, external/autonomous actions, overrides, DNC/suppression, connector/sending/model-routing/learning-policy changes, privileged actions, failures and manual interventions shall be auditable.

74. Compliance Policy Layer

Compliance-sensitive execution shall not rely solely on model judgment. Backend policies shall enforce applicable channel, Campaign, geography/country, suppression, authority, contact and other approved operational rules. The implementation shall support the separately developed AI Sales Outbound / Revenue Engine Privacy & Compliance Framework.

75. Security & Data Minimisation

Models/connectors shall receive only data reasonably necessary for authorised tasks. Credentials shall never be embedded in prompts, frontend/client code or ordinary workflow data. Secrets shall use secure configuration/secrets architecture. AI processing shall not itself expand access to production prospect/client data.

76. Reliability & Failure Semantics

Failures must remain truthful: Research failure != Disqualified; Connector failure != Sent; Provider failure != Rejected Prospect; Scheduling failure != Booked. Partial failure shall identify the failed step. Failure of one source may trigger another permitted source but may not silently alter policy or channel authority.

77. Idempotency

Consequential actions shall be idempotent where required. Retries, webhooks, queue replay or UI resubmission shall not create duplicate emails, submissions, follow-ups, scheduling requests or consequential opportunity actions.

78. Durable Background Processing

Large imports, research operations and Campaigns shall use durable background processing with progress, controlled concurrency, retry, pause/resume where applicable, recovery, rate-limit handling and failure visibility. A browser session shall not need to remain open.

79. Concurrency & Scale

Support multiple concurrent Campaigns, prospects, sources, research tasks, model operations and connector operations. Ten simultaneous prospects is not an architectural maximum. The system shall scale materially beyond launch volumes without fundamental redesign while respecting provider limits.

80. Configuration Ownership

Operational Product Owner decisions that require future adjustment shall not be unnecessarily hardcoded. Authorised configuration shall cover ICPs, Campaign Mandates, sources, connectors, channels,

communication profile, follow-up, authority matrix, sending identities, frequency controls, qualification, model routing, budgets, learning and geography as appropriate. Material configuration should support Draft -> Test/Preview -> Authorised Publish -> Active -> Version History/Rollback where appropriate.

81. Super Admin Control Center

Location: Super Admin -> AI Agents & IVR -> AI Sales Outbound. Provide coherent operational access to Overview, Sources, Organisations, People/Prospects, Signals, Evidence, Opportunities, Research, Qualification, Campaigns, Drafts, Approval Queue, Outreach, Replies, Follow-ups, Scheduling/Meeting Intent, Existing Client Opportunities, Human Attention, Imports, Connectors, Sending Infrastructure, Model Routing, Learning, Costs, Analytics, Failures, Configuration and Audit History. The objective is operational control, not simply a chat window.

82. Analytics

Minimum analytics include discovered, researched, qualified, disqualified, insufficient evidence, drafts, approved, edited, rejected, sent/submitted, failures, replies, positive replies, objections, DNC/unsubscribe, meetings, opportunities, Campaign outcomes, source/channel performance, model usage/cost and conversion. Commercial funnel: Discovered -> Researched -> Qualified -> Contacted -> Replied -> Interested -> Meeting -> Opportunity -> Client. Filter by source, Campaign, ICP, geography, channel, Signal type, pitch approach and period as relevant.

83. Learning Analytics

Expose meaningful patterns such as strong/weak sources, successful/weak approaches, common rejection reasons, frequent user corrections, model escalation rates, qualification performance and outcome trends. Analytics inform Controlled Learning; they do not independently rewrite governance.

84. Requirements Traceability

Every mandatory requirement shall be traceable to implementation and acceptance. Before acceptance, developers shall provide: Requirement ID -> Implemented Component -> Environment -> Test/Evidence -> Result -> Known Limitation. UI existence alone is not completion.

85. Acceptance Evidence

Evidence may include automated tests, integration tests, API evidence, browser/E2E tests, screenshots/video, state/database verification, audit records, provider/connector confirmations and failure-path demonstrations. Critical workflows shall be demonstrated in the deployed environment.

86. End-to-End Acceptance Scenario

Demonstrate: real Signal -> Evidence -> Organisation Resolution -> Relationship Check -> Research -> Qualification -> Contact Discovery -> Opportunity -> Personalised Communication -> Approval or Valid Autonomous Authority -> Real Authorised Channel Execution -> Confirmed Result -> Commercial Lifecycle Update. AI generated an email does not satisfy this requirement.

87. Source Acceptance

Multiple launch sources shall function end-to-end. Demonstrate different source types, real discovery, Evidence provenance, duplicate Signal handling and accurate connector capabilities. The same underlying opportunity from two sources shall not become two unrelated prospects.

88. Bulk Acceptance

Demonstrate realistic CSV/XLSX -> Validation -> Normalisation -> Deduplication -> CRM Match -> Relationship Check -> Research -> Qualification -> Recommended/Approval Actions at meaningful multi-record scale.

89. Commercial-Context Acceptance

Demonstrate permanent-employment vacancy, freelance/contract vacancy, different evidence-appropriate approach, previously contacted Organisation without new Signal, previously contacted Organisation with materially new Signal, Existing Client with new opportunity, Former Client and ambiguous Organisation identity.

90. Autonomy Acceptance

Demonstrate Research Only, Approval Mode, Autonomous Mode, Automatic action, Approval Required action, Prohibited action, Pause, Resume, Stop, Kill Switch and stale queued-action revalidation. The agent must remain within Campaign Mandate.

91. Reply & Scheduling Acceptance

Test Interested, More Information, Objection, Not Now, Wrong Contact, Referral, OOO, unsubscribe, Meeting Intent and complex Human Attention case. Demonstrate Meeting Intent -> Scheduling Agent -> booking and Call me Friday -> Scheduling Task with Sales Outbound context.

92. Model Routing Acceptance

Demonstrate low-complexity economical routing, standard routing, difficult/conflicting escalation, structured handoff preserving completed work, exceptional escalation where justified, unresolved highest-tier safe termination, technical failure distinct from intelligence escalation, bounded retry/escalation, usage/cost visibility and central model remapping without workflow rewrite.

93. Learning Acceptance

Demonstrate that repeated authorised edits/rejections produce appropriate improvement in comparable future work while preserving global IO SKY rules, Campaign authority, suppression and permissions, and remaining explainable/correctable.

94. Negative-Path Acceptance

- DNC active -> send blocked
- Campaign stopped -> send blocked
- Action outside Mandate -> blocked

- Unauthorised role -> backend denial
- Ambiguous Organisation -> no blind outreach
- Existing Client -> no cold sequence
- Material change after approval -> approval invalidated
- Duplicate queued execution -> duplicate prevented
- Insufficient Evidence -> no fabricated fact
- Connector failure -> no false Sent
- Campaign collision -> controlled handling
- Unresolved highest permitted model -> Human Attention/Incomplete state

95. Connector Definition of Done

For each launch connector, deliver integration method, authentication, Capability Matrix, limitations, rate limits, data handled, failure handling, approval requirements, production prerequisites and acceptance evidence. A connector is not complete merely because its name appears in the UI.

96. Cross-Specification Authority

Specialist specifications remain authoritative for their domains: Scheduling -> IO SKY Scheduling Agent Functional Specification; inbound AI calls -> AI Receptionist Functional Specification; Receptionist dialogue -> AI Receptionist Conversation & Response Playbook; IVR/telephony foundation -> IVR & Telephony Functional Specification. AI Sales Outbound remains authoritative for commercial discovery, research, qualification, Campaign authority, outbound strategy and commercial handoffs.

97. Developer Implementation Freedom

Developers remain responsible for appropriate database implementation, queues, service boundaries, APIs, caching, indexing, retry mechanisms, infrastructure, deployment and observability, provided all functional, security, reliability, authority and acceptance requirements are satisfied. Technical freedom is not permission to reduce scope.

98. Requirement Conflict / Dependency Rule

If a requirement is technically contradictory, impossible with an actual provider, dependent on unavailable external functionality, materially insecure/non-compliant or inconsistent with another authoritative IO SKY specification, the team shall raise: affected requirement -> technical reason -> impact -> proposed options. It shall not be silently omitted, weakened or reinterpreted.

99. No Requirement by Implication

Developers are not expected to predict functionality genuinely absent from the specification. Conversely, an explicitly present requirement may not later be treated as out of scope merely because its technical implementation was more extensive than anticipated.

100. Final Authority Architecture

Control hierarchy: IO SKY / Super Admin -> Platform Security, Permissions & Policy -> Campaign Mandate -> AI Research, Reasoning & Planning -> Backend Revalidation -> Authorised Connector / Action Layer -> Confirmed External Result -> Commercial Administration + Audit + Analytics -> Controlled Learning.

The AI may operate with substantial autonomy. It may not enlarge its own authority envelope.

101. Final Definition of Done

AI Sales Outbound v1.0 is not functionally complete until this lifecycle operates coherently: Discovery -> Signal Detection -> Organisation Resolution -> Relationship Protection -> Research -> Evidence -> Qualification -> Contact Discovery -> Opportunity -> Personalisation -> Approval / Authorised Autonomy -> External Execution -> Reply -> Follow-up -> Scheduling -> Commercial Administration -> Analytics -> Controlled Learning.

The following alone are insufficient: AI generates emails; AI searches vacancies; source cards exist; prospects can be imported; messages can be copied manually; chatbot responds to sales questions; mocked demo succeeds; autonomous button exists; one happy-path email is sent. Completion requires the integrated operational system specified above.

Final Specification Principles

Research before outreach.

Evidence before assumption.

Qualification before automation.

One organisation, one coherent commercial history.

A new Signal may create a new Opportunity; it never erases previous context.

Existing clients are protected from cold prospecting.

Autonomy exists within authority.

AI reasoning never replaces backend permission.

Use the least expensive model capable of meeting required quality; escalate through structured handoff when necessary.

Learning improves execution but cannot silently rewrite IO SKY governance.

Every consequential external action must be authorised, attributable and auditable.

Every automated workflow must leave IO SKY's administration more complete and organised, never more fragmented.

Commercial quality and operational efficiency take priority over maximum outbound volume.

Appendix A - Requirement Traceability Index

The following explicit requirement IDs are authoritative anchors. Sections without a dedicated ID remain mandatory where written with 'shall', 'must' or equivalent mandatory language.

Requirement ID	Authoritative Section	Requirement
ASO-CORE-001	Section 1 - Product Definition & Objective	AI Sales Outbound shall automate commercial execution without automating away IO SKY's control.
ASO-CORE-002	Section 1 - Product Definition & Objective	Every automated workflow must leave IO SKY's commercial administration more complete and organised, never more fragmented.
ASO-CORE-003	Section 1 - Product Definition & Objective	The objective is higher-quality commercial opportunities with substantially less manual work, not maximum outreach volume.
ASO-ARC-001	Section 2 - System Architecture & Authority	Critical business state shall be stored outside the LLM in authoritative platform/database services.
ASO-ARC-002	Section 2 - System Architecture & Authority	Where AI reasoning conflicts with authoritative platform state, authoritative platform state wins.
ASO-AUT-001	Section 3 - Authority Chain	Failure of any required authority condition shall block execution.
ASO-AUT-002	Section 3 - Authority Chain	The AI may never grant itself additional authority.
ASO-AUT-003	Section 3 - Authority Chain	Model intent is not permission; backend validation remains authoritative.
ASO-AUT-004	Section 4 - Operating Modes	Autonomous Mode means independent execution inside an explicitly controlled authority envelope; it never means unrestricted AI authority.
ASO-AUT-005	Section 6 - Campaign Lifecycle	Resume shall never blindly execute stale queued actions; current-state validation must occur first.
ASO-AUT-006	Section 7 - Pre-Action Revalidation	A previously valid decision does not override newer authoritative state. Example: approved draft -> unsubscribe -> queued send must be blocked.
ASO-DATA-001	Section 10 - Organisation Identity Resolution	Materially ambiguous identity shall not be blindly merged; route to review/Human Attention where necessary.
ASO-DATA-002	Section 12 - Signal	A Signal is not proof that IO SKY is needed; it triggers investigation.
ASO-RES-001	Section 13 - Evidence	Verified Fact, Evidence, Inference/Hypothesis and Unknown may never be silently collapsed.
ASO-CON-001	Section 23 - Connector Capability Matrix	A visible source logo/card is not a completed integration.
ASO-RES-002	Section 26 - Employment-Type Intelligence	Permanent-employment and freelance/contract situations shall not simply receive the same template with different job titles.
ASO-RES-003	Section 32 - Research Integrity	No invented problems: hypotheses may not become facts.
ASO-RES-004	Section 32 - Research Integrity	Time-sensitive evidence shall be freshness-aware.

ASO-RES-005	Section 32 - Research Integrity	Conflicting evidence shall trigger further research, stronger-model escalation, unresolved state or Human Attention - not fabricated certainty.
ASO-RES-006	Section 32 - Research Integrity	Research failure shall produce Research Incomplete / Insufficient Evidence, not Disqualified unless independent evidence justifies disqualification.
ASO-QUAL-001	Section 36 - Service Relevance	The existence of an IO SKY service is not evidence that a prospect needs it.

Appendix B - Developer Acceptance Matrix Template

This matrix shall be completed by the development team before final acceptance. Add rows for every mandatory requirement and acceptance scenario.

Requirement / Section	Implemented Component	Environment	Test / Evidence	Result	Known Limitation

Developer Handoff Rule

If a requirement cannot be implemented as specified because of a real technical, provider, security, compliance or cross-specification dependency, raise it to IO SKY with the affected requirement, reason, impact and proposed options. Do not silently omit, weaken or reinterpret it.